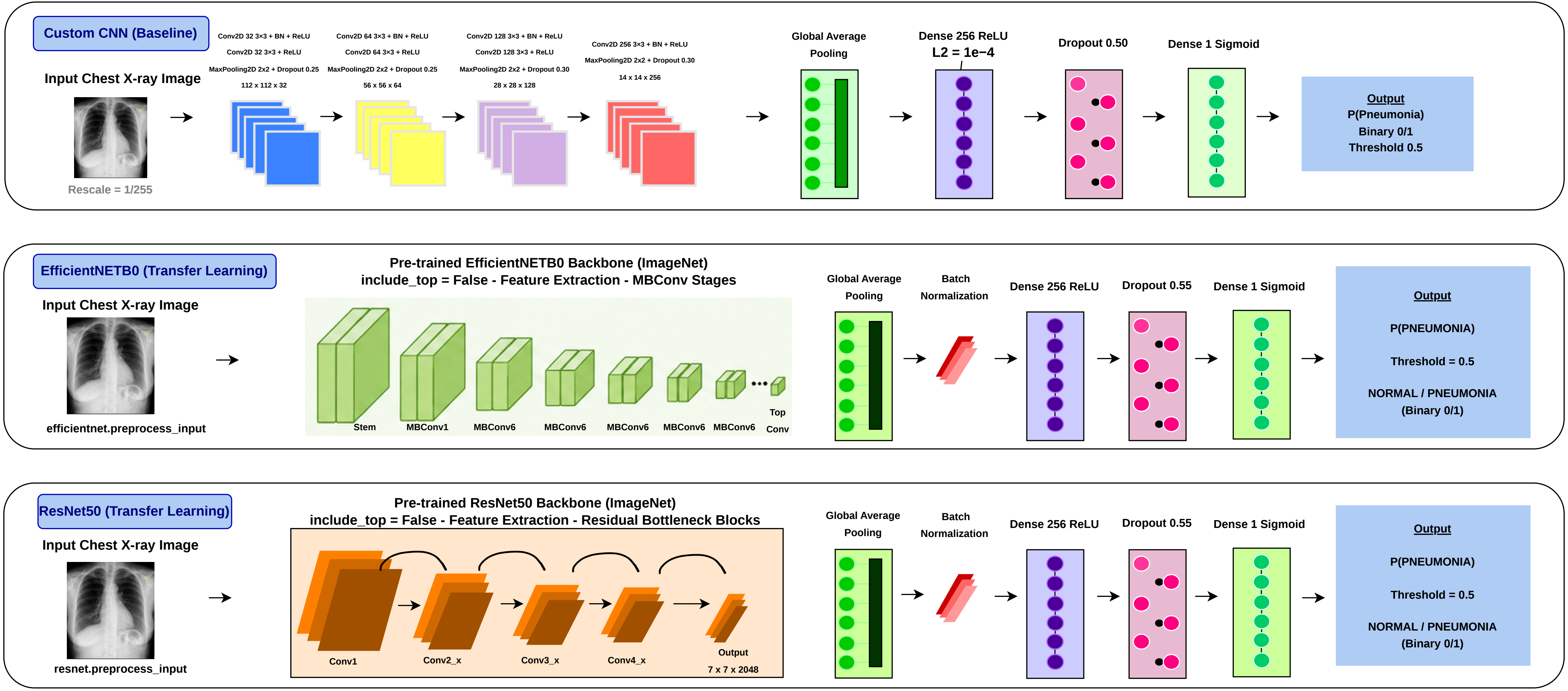
Custom CNN (Baseline)
Input Chest X-ray Image
Rescale = 1/255
Conv2D 32 3×3 + BN + ReLU
Conv2D 32 3×3 + ReLU
MaxPooling2D 2x2 + Dropout 0.25
112 x 112 x 32
Conv2D 64 3×3 + BN + ReLU
Conv2D 64 3×3 + ReLU
MaxPooling2D 2x2 + Dropout 0.25
56 x 56 x 64
Conv2D 128 3×3 + BN + ReLU
Conv2D 128 3×3 + ReLU
MaxPooling2D 2x2 + Dropout 0.30
28 x 28 x 128
Conv2D 256 3×3 + BN + ReLU
MaxPooling2D 2x2 + Dropout 0.30
14 x 14 x 256
Global Average Pooling
Dense 256 ReLU
L2 = 1e−4
Dropout 0.50
Dense 1 Sigmoid
Output
P(Pneumonia)
Binary 0/1
Threshold 0.5
EfficientNETB0 (Transfer Learning)
Input Chest X-ray Image
efficientnet.preprocess_input
Pre-trained EfficientNETB0 Backbone (ImageNet)
include_top = False - Feature Extraction - MBConv Stages
Stem
MBConv1
MBConv6
MBConv6
MBConv6
MBConv6
MBConv6
Top Conv
Global Average Pooling
Batch Normalization
Dense 256 ReLU
Dropout 0.55
Dense 1 Sigmoid
Output
P(PNEUMONIA)
Threshold = 0.5
NORMAL / PNEUMONIA
(Binary 0/1)
ResNet50 (Transfer Learning)
Input Chest X-ray Image
resnet.preprocess_input
Pre-trained ResNet50 Backbone (ImageNet)
include_top = False - Feature Extraction - Residual Bottleneck Blocks
Conv1
Conv2_x
Conv3_x
Conv4_x
Output
7 x 7 x 2048
Global Average Pooling
Batch Normalization
Dense 256 ReLU
Dropout 0.55
Dense 1 Sigmoid
Output
P(PNEUMONIA)
Threshold = 0.5
NORMAL / PNEUMONIA
(Binary 0/1)